RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai)
Thandalam, Chennai – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS
CASE STUDY ASSIGNMENTS – II

CONSOLIDATED SUBMISSION
(All Five Case Studies – Questions 1 to 38)

CANDIDATE AND COURSE SUBMISSION PARTICULARS	
Student Name	SREE DHARSHINI M
Register Number	2116242901110
Degree / Department	B.E. CSE – Cyber Security
Year / Semester	3rd Year / 5th Semester
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Faculty In Charge	Dr. Sivakumar D
Institution	Rajalakshmi Engineering College
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CONTENTS OF THE CONSOLIDATED SUBMISSION

Each case study follows the same structure — case overview with facts separated from assumptions, the numbered assignment questions with their deliverable tables and figures, limitations, conclusion and references. All case data is hypothetical and is labelled as such; no forensic tool was executed for this coursework.

Case Study	Title	Questions	Scenario ID
2(1)	Suspected Unauthorized Access to a College Server	Questions 1–7	REC-AMPL-2026-01
2(2)	Linking a Suspicious File to a User Using File-System Evidence	Questions 8–14	MER-FIN-2026-33
2(3)	Forensic Imaging and Safekeeping of a Suspect Storage Device	Questions 15–22	CT-2026-114
2(4)	Recovery of Deleted Evidence from Volume Shadow Copies	Questions 23–30	DGN-2026-07
2(5)	Multi-Source Investigation of a Suspected Cyber Incident	Questions 31–38	IDO-2026-19

Page map of this consolidated PDF

Master cover and contents — pages 1–2
Case Study 2(1) — Questions 1–7 — pages 3–12
Case Study 2(2) — Questions 8–14 — pages 13–22
Case Study 2(3) — Questions 15–22 — pages 23–32
Case Study 2(4) — Questions 23–30 — pages 33–42
Case Study 2(5) — Questions 31–38 — pages 43–52

Note on formatting: one front page serves the whole consolidated submission — the college logo and the candidate particulars appear once, here at the top. Each case study then starts on a new page, tagged in the top margin of its opening page, and keeps its own footer page numbering (2, 3, …) so the five reports remain independently referenceable against the separate submissions.

ABSTRACT

This report examines a hypothetical incident at the Advanced Materials & Prototyping Laboratory (AMPL), a college-affiliated research laboratory, in which five confidential prototype-design documents were suspected of having been copied from the departmental file server to laboratory removable media outside working hours. It answers Questions 1 to 7 and produces the four required deliverables: an investigation flowchart, a classified evidence table, the case analysis and a sample forensic report structure, following the five-phase model of NIST SP 800-86 and ISO/IEC 27037. The artefacts show that the folder was copied; they do not identify who copied it, because the login, the entry card and the memory card were all shared — a limit reported as a result rather than concealed. No tool was executed for this coursework; case values are scenario data.

CONTENTS

1.	Case background, known facts and assumptions
2.	Detailed answers to Questions 1 to 7
3.	Case analysis, findings and limitations
4.	Conclusion
5.	References

1. CASE BACKGROUND, KNOWN FACTS AND ASSUMPTIONS

Dr. N. Bhuvaneswari, laboratory coordinator at AMPL, reported that the sponsored project folder ProtoX_RevC on file server NAS-PRJ-02 had been read and copied on the evening of 3 October 2026. The folder is edited from one bench workstation, AMPL-WS07, under two shared laboratory logins. The next morning a 128 GB microSD card, normally kept in a reader in the same drawer, was found to hold copies of five project documents. Because the sponsor's confidentiality terms apply, the institution constituted a team to establish what happened without contaminating the original evidence.

Table 1.1: Known Facts, Hypothetical Scenario Data and Analytical Assumptions

Category	Content and how it is used
Given by the assignment brief	Administrator complaint; confidential files copied outside working hours from a departmental server; an evidence set of complaint, suspect workstation, server logs, a recovered device and a few files; the duty not to contaminate originals.
Hypothetical scenario data	Case id REC-AMPL-2026-01; AMPL-WS07 (Windows 11 Pro, NTFS, 512 GB NVMe); NAS-PRJ-02 (Linux, ext4 over SMB); share \\NAS-PRJ-02\ProtoX_RevC; shared logins ampl_ops and ampl_student1; window 3 Oct 2026, 20:05–20:34 IST; files ProtoX_Housing_revC.step, Enclosure_Tolerance_Matrix.xlsx, Thermal_Test_Log.docx, BOM_ProtoX_revC.csv, Sponsor_Confidentiality_Terms.pdf; 128 GB exFAT microSD in a generic USB reader.
Analytical assumptions	Written authorisation from the Dean (Research) preceded acquisition; NAS log rotation frozen at intake; workstation found powered on, so volatile data taken first; NAS clock NTP-locked, workstation clock free-running with a measurable offset.
Out of scope	Naming an individual, motive and any disciplinary decision — these rest with the committee and need corroboration outside the digital artefacts, since the login, the badge and the card were shared.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 1–7)

2.1 Movement Through the Five Phases (Question 1)

> **Question 1: Explain how the forensic team should move through the Identification, Collection, Examination, Analysis, and Presentation phases for this case.**

Each phase is closed only when an output exists that the next phase can consume. The discipline matters more here than in a corporate case: shared equipment means a spoiled acquisition can neither be repeated nor explained away to a committee.

Table 1.2: Five Phases Applied to the AMPL Incident

Phase	Action in this case	Closing output
Identification	Record the complaint and the reported window; obtain the written mandate; scope AMPL-WS07, NAS-PRJ-02 and the card; list corroborating sources (door log, backup schedule, sponsor retention terms).	Authorisation memo; scoped intake form; source inventory
Collection	Photograph bench, reader and card in place; measure clock offsets; capture memory, process list and open handles from the live workstation; isolate; write-blocked images of volume and card; read-only export of NAS logs.	Sealed exhibits AMPL-E1 to E4; images with digests; custody register
Examination	On verified read-only copies: $MFT, $UsnJrnl and Security.evtx; USBSTOR, MountPoints2, DeviceTree; the exFAT directory and its deleted-entry slots; NSRL filtering to keep review on the five documents.	Artefact tables and file listings, never from originals
Analysis	Normalise times to UTC with IST shown; align logon, connect, read, mount and write events; compare digests of card copies against share originals; test the benign backup explanation.	Master timeline; hypothesis evaluation; attribution position
Presentation	Layered report — plain summary, technical annexures, hash and custody appendices — briefed to the committee with an explicit statement of what remains unproven.	Final report; briefing record; archived package

2.2 Investigation Flowchart: Complaint to Presentation (Question 2)

Question 2: Prepare a forensic investigation flowchart showing the sequence of activities from the initial complaint to presentation of findings.

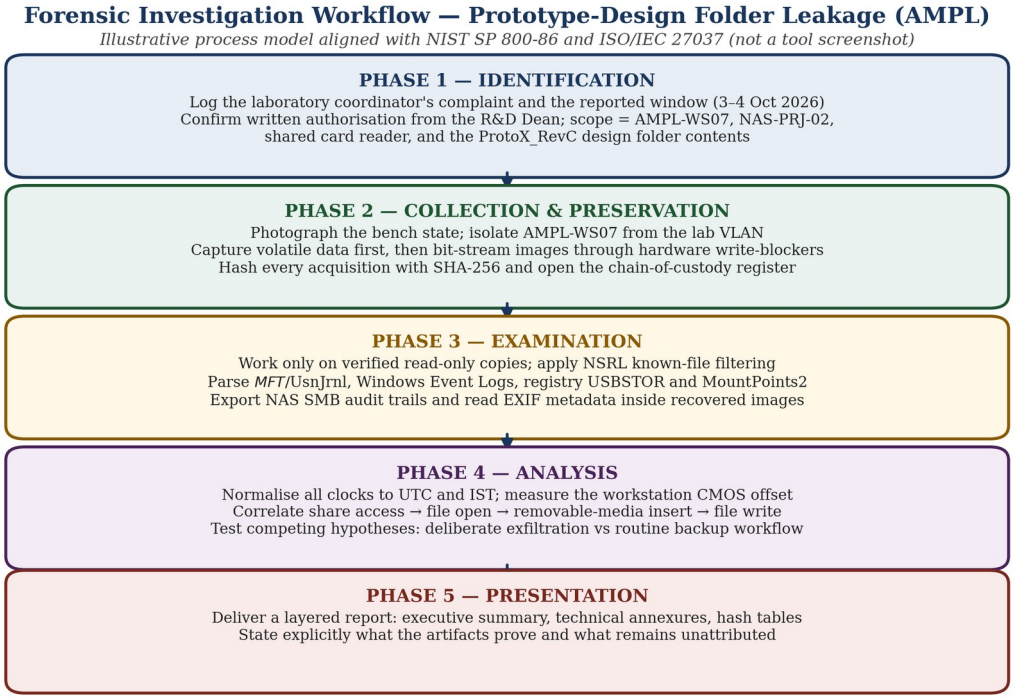


Figure 1.1: Investigation flowchart for the prototype-folder incident (original illustrative diagram, not a tool screenshot)

Figure 1.1 runs top to bottom with a gate between phases, and two gates decide this case. Before Phase 2 the workstation is untouched until memory and logon state are recorded, because the session that matters was still open when the card was found. Before Phase 4, working copies are re-hashed, so examiner tool activity can never be read as case activity.

2.3 Potential Digital Evidence and Classification (Question 3)

Question 3: Identify at least five items of potential digital evidence in the case and classify the evidence according to its investigative value.

Table 1.3: Evidence Inventory Classified by Investigative Value

Item (exhibit)	Form and location	Class	Value in this case
Coordinator's complaint	E-mail and incident note	Originating / contextual	Fixes report and discovery times, bounding the search window; not proof of copying
AMPL-WS07 volume image (E2)	NTFS; logon, access, mount records	Primary system evidence	Strongest item: logon, connect, device arrival and writes on one machine

Item (exhibit)	Form and location	Class	Value in this case
NAS audit log export (E3)	SMB and authentication lines	Primary system evidence	Independent second clock for the five reads; exposes the benign backup job
microSD card and reader (E4)	exFAT; five copies, unallocated space	Recovered / derivative	Holds the outcome; exFAT times weak, so contents and digests carry it
The five project documents	Share originals and card copies	Subject-matter evidence	Digest identity turns suspicion of copying into a demonstrated copy
Registry shell and device keys	USBSTOR, MountPoints2, UserAssist	System-generated	Binds this serial to this machine and profile, making the writes explicable

Classification follows what an item can carry, not how technical it looks; the door-controller record was therefore treated as corroboration rather than case evidence, being an estates access log outside the mandate.

2.4 Precautions During Collection and Preservation (Question 4)

> **Question 4: Explain the precautions required during collection and preservation so that the evidence remains reliable and defensible.**

In a teaching laboratory the failure modes are casual rather than adversarial — a student replugs the reader to check the card, a technician reboots to install a driver. Each precaution closes one such gap.

- **Authority before access.** Acquisition starts only after the written mandate, which also names the custodian entitled to export NAS logs.
- **Order of volatility (RFC 3227).** Memory, processes, open handles and network sessions precede isolation and power-down.
- **Two layers of write protection.** Hardware blocker per source plus read-only mounting of copies; the SD lock tab checked and recorded.
- **Hash at first contact and at each hand-off.** Source hashed before copying and image after, both entered in the register, so later change is detectable.

- **Originals sealed; offsets recorded.** Copies are analysed so a mistaken parser run costs nothing, and clock offsets are measured at intake, since a few unrecorded minutes can make a mount appear to precede the reads that caused it.

2.5 Establishing a Timeline of the Suspected Access (Question 5)

Question 5: Describe how investigators can establish a timeline of the suspected unauthorized access using available evidence.

The timeline is assembled from three independent clocks before it is narrated: the workstation, which holds both the logon and the mount record; the NAS, which confirms the reads on another clock; and the card, the weakest witness because exFAT stores dates without a time zone. Figure 1.2 draws the three lanes and the event register that the table below then grades.

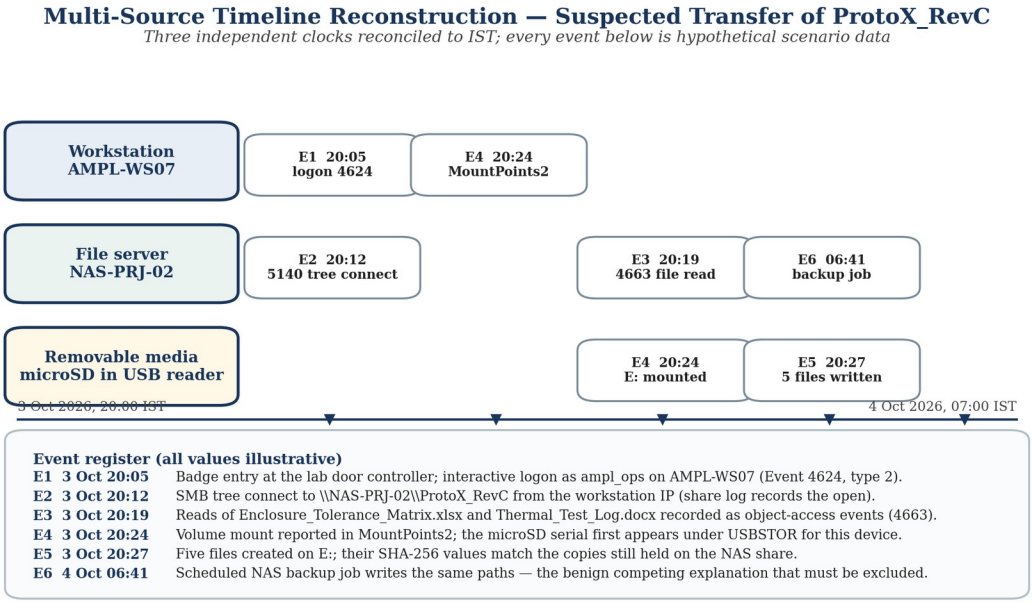


Figure 1.2: Multi-source timeline reconstruction with event register (illustrative scenario data)

Table 1.4: Correlation and Strength of Each Reconstructed Event

Event	Source artefact	Establishes	Strength
E1 20:05 logon as ampl_ops	4624 type 2, Security.evtx	Console session; activity locally initiated	Strong for session, silent on person
E2 20:12 share connect	SMB log, Event 5140	Folder opened from that machine that minute	Strong, independent clock
E3 20:19 five reads	NAS object-access audit	Named confidential files, not just the folder	Strong

Event	Source artefact	Establishes	Strength
E4 20:24 card mount	MountPoints2, USBSTOR	This serial appeared between E3 and E5	Strong device link
E5 20:27 five writes	exFAT entries; matching digests	Share contents reproduced on removable media	Strong for the act
E6 06:41 backup job	NAS scheduled-task log	Automation also touches those paths	Exculpatory until excluded

An event enters the timeline only when a second source places it in the same minute; otherwise it is reported as inference. E1 to E5 form a chain no documented laboratory workflow reproduces, while E6 shows why timing alone cannot label a copy unauthorised.

2.6 Final Report Contents and Presentation to a Non-Technical Authority (Question 6)

Question 6: Explain what should be included in the final forensic report and how findings should be presented to a non-technical authority.

Table 1.5: Structure of the Sample Forensic Report

Section	Content for this incident
1. Executive summary	Four sentences: what was reported, what the artefacts show, what they do not show, and the recommended control response.
2. Scope and authority	The mandate, its signatory, exhibits covered and exclusions agreed at intake.
3. Background, evidence and method	Laboratory and shared-login context; exhibit table with acquisition times and digests; tools, versions, write-blocking and clock measurements.
4. Findings	Numbered, each naming the artefact behind it and graded established, supported or unresolved.
5. Limitations and appendices	What remains unattributed; artefact extracts, hash lists, custody forms and a glossary.

For a non-technical audience the rule is one assertion, one exhibit, one plain sentence: five documents were copied to the laboratory's own card between 20:19 and 20:27 on 3 October and are identical to the server versions — followed by the exhibit that shows it. Terms such as $MFT stay in annexures, no paragraph names a person, and every finding carries a grade.

A committee given graded findings can act; one given an accusation inherits a burden of proof it cannot carry.

2.7 Legal and Ethical Considerations in a College Environment (Question 7)

Question 7: Briefly discuss the legal and ethical considerations involved in collecting digital evidence in a college environment.

Table 1.6: Considerations Binding the Team

Consideration	Instrument	Requirement here
Lawful basis for access	Institutional IT policy; enrolment terms	Institution-owned equipment inside the mandated scope only; personal devices need separate consent.
Personal data minimisation	Digital Personal Data Protection Act, 2023	Identities minimised; only the extract bearing on a finding enters the report body.
Integrity and admissibility	IT Act, 2000, ss. 43, 66; Bharatiya Sakshya Adhiniyam, 2023, s. 63	Hashing, custody records and contemporaneous notes let an electronic record be shown secure.
Third-party confidentiality	Sponsor and non-disclosure terms	Design content is not reproduced; identifiers and digests are used instead.
Fairness and proportionality	Disciplinary procedure; research ethics practice	Persons named see the assertion with its artefact basis and may respond; review stops at the window.

The recurring ethical pressure in laboratory cases is scope creep: an examiner can technically read an entire shared profile, and might even find something. The mandate sets the boundary, and the committee's ability to rely on the result depends on that boundary being respected.

3. CONSOLIDATED CASE ANALYSIS, FINDINGS AND LIMITATIONS

Table 1.7: Hypotheses Tested Against the Artefacts

Hypothesis	Supports it	Contradicts it	Position
H1 A logged-in user deliberately copied the folder to removable media	E1–E5 sequence; five matching digests	None in the window	Supported for the act; actor unresolved
H2 An automated process staged the files	E6 shows automation touching those paths	It writes only to the backup target; no job at 20:27	Rejected for card writes; caveat retained
H3 The card already held these files from an earlier transfer	exFAT dates are weak alone	This serial first appears in USBSTOR at E4, after E3	Rejected on device registration
H4 A remote session produced the activity	Explains a login with nobody at the desk	4624 records type 2 console; no remote logon	Not supported; remote services to be listed

Three findings survive. Five documents were read from the share and written to laboratory microSD media within a twenty-two minute window opened by a console logon; the copies are byte-identical to the share versions, which forecloses the unrelated-draft argument; and nothing in the examined set identifies an individual, because the login, the badge and the card were shared and no CCTV record was available.

Table 1.8: Limitations and What Would Lift Them

Limitation	Effect	Corroboration required
Shared laboratory credentials	Establish a group, never a person	Door-controller name map, bench CCTV, roster of card holders
exFAT timestamp weakness	Card times cannot sequence alone	Workstation mount and write records, as used at E4–E5
Free-running workstation clock	Sub-minute order uncertain	Recorded offset against the NTP-locked NAS
Text log export from the NAS	Editing cannot be excluded	Forwarded syslog or independently signed archive

4. CONCLUSION

This incident yields a chain of events that is checkable rather than merely plausible: a console session, an independent server record of the confidential reads, a device registration tying one

microSD card to that machine and minute, and five files whose digests match the share copies. The evidence shows that the prototype folder left the laboratory network on laboratory media, and shows it against the project team as a whole rather than against an individual. That asymmetry is the consequential line here, because the response it points to is control rather than accusation: per-user credentials on shared research equipment, removal of idle write access to sponsor folders, port-level management of removable media, and a written transfer procedure for cards leaving a bench.

5. REFERENCES

[1] Kent, K., Chevalier, S., Grance, T., & Dang, H. (2006). NIST SP 800-86: Guide to Integrating Forensic Techniques into Incident Response. NIST.

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[3] Brezinski, D., & Killalea, T. (2002). RFC 3227: Guidelines for Evidence Collection and Archiving. IETF.

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[6] Government of India. (2000). The Information Technology Act, 2000 (as amended in 2008), Sections 43 and 66; (2023) The Bharatiya Sakshya Adhiniyam, 2023, Section 63; Digital Personal Data Protection Act, 2023.

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ABSTRACT

This report answers Questions 8 to 14 of Assignment 2(2) on a hypothetical examination of FIN-WS-12, a finance workstation shared by three staff members and one service account, on which an unfamiliar PowerShell script, a vendor-payment exception spreadsheet, hidden synchronisation configuration and an encrypted archive were found. It supplies the four required deliverables: a documented hardware and software inventory, an evidence-linkage matrix, a conceptual import-and-analysis workflow in ILookIX, and a conclusion written with its limitations attached. The analytical interest here is the scheduled task that runs the same script nightly as a service account: it turns attribution from a search for a name into a test of logon type, token ownership and profile scope. No software was executed for this coursework; the workflow is described from syllabus concepts and every value is scenario data.

CONTENTS

1.	Case background, facts and assumptions
2.	Detailed answers to Questions 8 to 14
3.	Conclusion with limitations
4.	References

1. CASE BACKGROUND, KNOWN FACTS AND ASSUMPTIONS

Meridian Foods Pvt Ltd keeps one Windows workstation, FIN-WS-12, for accounts-payable work at the college-affiliated incubator office it shares with a faculty project team. A clerk reported an unfamiliar desktop entry named recon_export.ps1 and a new file, Vendor_Payment_Exceptions_Q3.xlsx, in the finance folder. The department's payroll recalculation archive also appeared locked on the morning of 6 October 2026. Three staff profiles and one service account use this machine. Investigators are asked to examine the environment and to link the files, if the artefacts permit, to one account.

Table 2.1: Known Facts, Hypothetical Scenario Data and Assumptions

Category	Content
Given by the brief	A desktop used by several staff members; a suspicious executable-class item; hidden files; encrypted documents; browser-related artefacts; multiple user accounts; a suspicion that one account created and accessed the files; instruction to examine the environment and analyse with a tool such as ILookIX.
Hypothetical scenario data	Case id MER-FIN-2026-33; FIN-WS-12 (Intel Core i5-1135G7, 16 GB RAM, 512 GB NVMe, BitLocker on drive C:); Windows 11 Pro 23H2, NTFS with ReFS on a data volume; profiles mfr_ap, mfr_admin and mfr_payroll_svc; hidden C:\ProgramData\SyncAgent\conn.cfg and sync_profile.json; encrypted Payroll_Recalc_Archive.zip; Task Scheduler entry FinanceSync, daily 02:00 as mfr_payroll_svc.
Assumptions	The employer owns the equipment and consented in writing; the archive is a company file, not personal data of an employee; volume shadow copies exist on C:; system and application log retention had not yet rolled over at acquisition.
Out of scope	Network capture at the gateway, review of the other two workstations, and any employment-law determination about misconduct.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 8–14)

2.1 Hardware and Software Components to Document Before Analysis (Question 8)

> **Question 8: Identify the hardware and software components that should be documented before analysis.**

Documentation precedes acquisition because later the analyst cannot tell what the machine was doing from what it became. The inventory below is the record an examiner completes before a blocker is attached, and each line exists to answer a question the defence or the HR committee will ask.

Table 2.2: Pre-Analysis System Inventory for FIN-WS-12

Component	Detail to record	Why it matters here
Chassis, serial and location	Dell OptiPlex 7090, service tag, tower, desk position	Ties the image to one physical exhibit (FIN-E1)
CPU, memory and firmware	i5-1135G7, 16 GB, BIOS version and date	Sets expectations for volatile capture and log granularity
Storage and encryption state	512 GB NVMe (GPT/NTFS) plus 1 TB ReFS data volume; BitLocker protectors and key custody	Decides what is imaged, in what order, and whether content is plaintext
Removable and attached media	USB device history; SD reader present or absent	Shows whether files could have left the machine
Operating system and build	Windows 11 Pro 23H2, install date, last update	Fixes the artefact set and time zones in play
Accounts and group membership	Three local profiles, one service account, local-admin members	The link table needs SIDs, not display names
Services, tasks and autostart	FinanceSync task, sync-agent service, startup entries	The competing automation explanation lives here
Security tooling and log policy	EDR agent, audit policy, log sizes and retention	Predicts which events survived to acquisition
Timekeeping and applications	NTP source and measured offset; Excel, PowerShell script-block logging, backup client	Clocks make correlation possible; applications explain metadata producers

2.2 Storage, Operating System, File System and Metadata in Locating Evidence (Question 9)

Question 9: Explain how storage devices, operating systems, file systems, and metadata can help investigators locate relevant evidence.

Table 2.3: What Each Layer Contributes in This Case

Layer	Contribution to locating evidence
Storage device	The NVMe volume holds everything the profiles touch; its allocation state tells the examiner whether the spreadsheet is live, deleted, or both, and the external ReFS volume holds the archive copy.
Operating system	Windows records sessions and objects separately: Security.evtx carries logon type, Sysmon or script-block logging carries command text, and each profile owns its own artefact namespace.

Layer	Contribution to locating evidence
File system	NTFS $MFT supplies created, modified, accessed and entry-modified times plus the owning SID; $UsnJrnl shows the sequence of create, rename and extend; $I30 index fragments reveal directory-listing times.
File metadata	Office documents carry Author, LastSavedBy, revision and TotalEditingTime; ZIP entries carry their own modified times, which frequently disagree with the file-system view and are not errors.
Registry	USBSTOR and MountPoints2 prove a device's presence; Run keys and the Task Cache prove what the system was told to do without a user.

In practice the layers are read inside out: file-system records choose which files deserve attention, metadata says who touched them, and the operating-system log decides whether a human was present at all.

2.3 Hidden and Encrypted Files and Their Handling (Question 10)

Question 10: Describe how hidden files and encrypted files may affect a forensic investigation. Discuss appropriate evidence-handling considerations.

Two distinct problems arise. Hidden items change what a listing shows; encrypted items change what the bytes mean. Both can defeat a keyword-driven examination without leaving any sign that they have done so, which is why the workflow below treats them as expected case features rather than as anomalies.

Table 2.4: Handling Rules for the Concealed and the Sealed

Situation in this case	Indicator	Handling rule
Hidden configuration folder C:\ProgramData\SyncAgent	Directory and file attributes set; contents absent from a default Explorer view	Enumerate with explicit hidden and system display; record attribute bits and the parent ACL alongside the path
Alternate data stream on the script	named stream on recon_export.ps1	Dump every stream separately and hash each; treat the primary stream and metadata stream as distinct exhibits
Encrypted archive Payroll_Recalc_Archive.zip	Local file headers present, data entries compressed and encrypted; high entropy	Preserve as-is; do not attempt recovery from a copy; note that modification times inside the archive are uncorroborated

Situation in this case	Indicator	Handling rule
BitLocker-protected system volume	Volume sealed at acquisition	Capture memory first to preserve unlocked keys, record protector custody, and document the recovery path sought from IT
Files only visible in a snapshot	Present in a shadow copy, absent from the live tree	Record which snapshot supplied which version, and hash each version separately

2.4 Evidence-Linkage Matrix (Question 11)

Question 11: Prepare an evidence-linkage table showing how file names, paths, timestamps, user accounts, metadata, and other artifacts could be correlated to a user.

Table 2.5: Evidence-Linkage Matrix for the Finance Workstation

Artefact	Path or location	Key times (IST)	Account / SID	Metadata link
recon_export.ps1	C:\Users\mfr_ap\Desktop	Created 03 Oct 21:14; modified 03 Oct 21:14; MFT entry 03 Oct 21:14	S-1-5-21-…-1004 (mfr_ap)	Owner SID equals the profile that holds the desktop path
Vendor_Payment_Exceptions_Q3.xlsx	\\MERFS01\Finance\Q3	Created 04 Oct 09:02; last saved 04 Oct 11:47	LastSavedBy = mfr_ap; owner = 1004	Author property, revision 7, TotalEditingTime 141 min
conn.cfg (hidden)	C:\ProgramData\SyncAgent	Created 02 Oct 02:00; modified 05 Oct 02:00	Owner S-1-5-21-…-1007 (mfr_payroll_svc)	Same timestamp as the FinanceSync task start
sync_profile.json	C:\ProgramData\SyncAgent	Modified 05 Oct 02:00:04	Owner mfr_payroll_svc	Contains the share path and a schedule id matching the task XML
Task Scheduler XML: FinanceSync	System32\Tasks	Trigger daily 02:00; last run 05 Oct 02:00	Runs as mfr_payroll_svc	Command argument points at recon_export.ps1
Browser download entry	History database, mfr_ap profile	03 Oct 21:13:52	mfr_ap profile only	Target URL, referrer and destination path matching the desktop file
Jumplist for Excel	…\Recent\CustomDestinations	04 Oct 11:47	mfr_ap	Bookmarks the workbook path with a volume id and file reference number

The matrix is read by row for corroboration and by column for independence: the download entry, the desktop file and the JumpList agree on mfr_ap within ninety seconds, while the ProgramData pair points at the service account and at 02:00, when nobody was logged on.

2.5 Import and Analysis Workflow in ILookIX (Question 12)

Question 12: Explain a suitable workflow for importing/acquiring the forensic evidence and analyzing it using ILookIX, based on the concepts covered in the syllabus.

ILookIX is used here as the syllabus describes it: an examination environment that imports an acquired image, rebuilds the file-system view including deleted entries, and presents partition, file and metadata structure to the analyst. The sequence below is a conceptual workflow written from those concepts; no case was processed and no output is reproduced in this report.

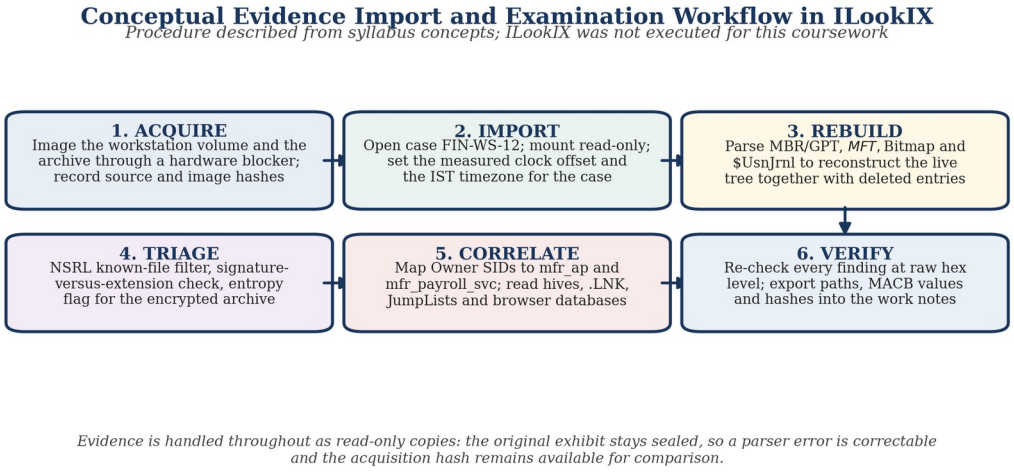


Figure 2.1: Conceptual ILookIX import and examination workflow (illustrative diagram, not a software screenshot)

Figure 2.1 shows the six stages in sequence. Stages 1 and 2 decide whether anything later is admissible, stages 3 and 4 decide whether the analyst is looking at the whole picture, and stages 5 and 6 are where attribution is either earned or lost.

- **Acquire, then import.** A write-blocked bit-stream image of the NVMe volume is taken, hashed, and the image — never the exhibit — is opened by the tool with a read-only mount.
- **Set the case clock, then rebuild.** The measured offset and IST zone are entered so exported times are normalised; partition, $MFT, $Bitmap and $UsnJrnl are parsed to

yield a live tree and a deleted tree, and the analyst records which is which for every file cited.

- **Rebuild structure.** Partition, $MFT, $Bitmap and $UsnJrnl are parsed to produce a live tree and a deleted tree; the analyst notes which is which for every file cited.
- **Reduce before reading.** Known-file filtering against an NSRL set removes operating-system and Office binaries, leaving a few hundred candidate artefacts.
- **Correlate, then verify at source.** Owner SIDs are mapped to profile names and hives, JumpLists, .LNK targets and browser databases are examined per profile rather than per machine; every finding is re-opened at hex level, with path, MACB times, SID and hash exported so a reviewer can repeat it.

2.6 Direct Linkage Against Supporting or Contextual Evidence (Question 13)

Question 13: Differentiate between an artifact that directly links a user to an activity and an artifact that merely provides supporting/contextual information.

Table 2.6: Grading the Artefacts in This Case

Artefact	Grade	Reason
Logon event, type 2, at 03 Oct 21:13 for mfr_ap	Direct	Places an interactive session of that account at the machine one minute before the script appeared
Download record in that profile's browser database	Direct	Names the destination path and account together, and its time matches the file creation
Owner SID of the desktop script	Direct for the profile, not the person	Proves which profile's namespace holds the file, not who typed
JumpList and RecentDocs entries	Supporting	Show a file was opened by that profile but carry no independent time of first creation
Excel LastSavedBy and revision metadata	Supporting	Application-written and editable; corroborates rather than proves
Script present in a shared folder	Contextual	Available to every profile; establishes opportunity only
FinanceSync task at 02:00 as service account	Exculpatory / alternative	Reproduces the writes without any user session, so it competes with the direct evidence

The test applied throughout is: would this artefact, standing alone and taken at face value, place an identified account at an identified action at an identified time? Anything failing that test is reported as supporting and is used to strengthen a direct link, never to substitute for a missing one.

2.7 Limitations and Alternative Explanations Before Attribution (Question 14)

Question 14: Identify at least three limitations or alternative explanations that investigators should consider before attributing the activity to a particular user.

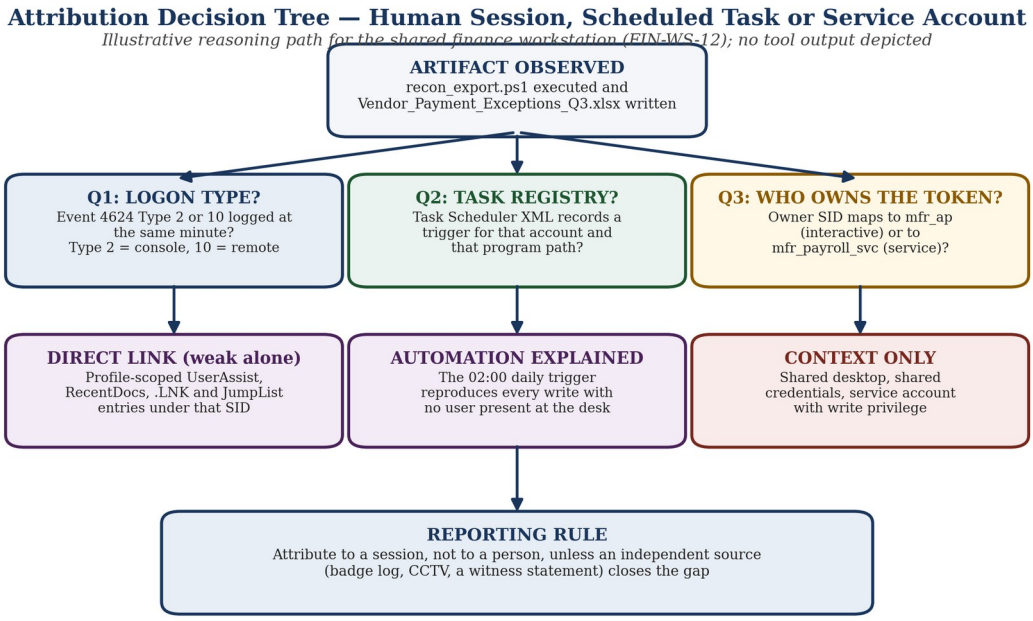


Figure 2.2: Attribution decision tree for the shared finance workstation (original illustrative diagram)

Figure 2.2 is the reasoning the limitations force: an artefact may be allowed to name a person only when a console or remote logon, the token owner and the absence of automation all agree.

Table 2.7: Limitations, Alternatives and the Test That Would Settle Each

Limitation or alternative	Why it bites here	Test or corroboration
L1 Shared machine, weak person-link	Three profiles and a service account use FIN-WS-12	Badge reader or CCTV at the office door for 03 Oct 21:00–21:20
L2 Automation explains the writes	FinanceSync runs the same script nightly as the service account	Compare task last-run time with each file's MFT times; disable and observe one cycle in a test image

Limitation or alternative	Why it bites here	Test or corroboration
L3 Inherited or copied timestamps	A relocated file keeps an older modified time, and ZIP entries keep theirs	Compare $UsnJrnl sequence with $MFT times; created-after-modified is a relocation marker
L4 Metadata is editable	LastSavedBy and Author are user-settable strings	Do not treat Office properties as identity; corroborate with file-system owner and logon records
L5 Clock skew across sources	Workstation free-runs; NAS is NTP-locked	Record and apply the measured offset before any ninety-second correlation is claimed
L6 Elevation and profile reuse	mfr_admin can write into another profile's namespace	Check for type-3 or remote logons and for privilege-use events around the write time

3. CONCLUSION WITH LIMITATIONS

On the hypothetical artefacts, the script and the workbook are linked to the mfr_ap interactive session of 3 October 2026 at 21:13–21:14 by three mutually independent sources — a type-2 logon event, a browser download record naming the destination path, and the owner SID of the created file — while the hidden synchronisation configuration and the payroll archive's recalculation belong to the service account and to the 02:00 scheduled trigger, with no human session in progress. The evidence therefore supports an attribution to a session, not to a person, and it explains part of the alarm as automation. Three limitations bound that conclusion: a shared machine cannot separate its three profile holders without physical or observational corroboration; scheduled automation reproduces writes that look deliberate; and Office metadata such as LastSavedBy is editable, so it corroborates but never identifies. The response these limits point to is credential and scheduling hygiene — individual accounts on finance equipment, script-block logging enabled, and service-account tasks documented, so that a nightly job is recognised in seconds rather than in a week of analysis.

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[6] Microsoft Corporation. (2024). NTFS technical reference: $MFT, $UsnJrnl and alternate data streams; Task Scheduler schema. Microsoft Learn.

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ABSTRACT

This report answers Questions 15 to 22 for a hypothetical seizure in which a laptop and a 64 GB camera memory card were taken from a distributor suspected of trading counterfeit bearings. The card had been formatted in the camera and is believed to hold product photographs, stock sheets and deleted shipment records. Four deliverables are produced: a collection and transport procedure with an acquisition flowchart, an imaging procedure built around IXImager, an integrity explanation based on hashing and the ASB container, and a completed chain-of-custody form. No tool was executed for this coursework; procedures are described from the syllabus and every value is scenario data.

CONTENTS

1.	Case background, facts and assumptions
2.	Detailed answers to Questions 15 to 22
3.	Consolidated integrity statement and conclusion
4.	References

1. CASE BACKGROUND, KNOWN FACTS AND ASSUMPTIONS

A warehouse office inspection on 6 October 2026 turned up packing slips for bearing assemblies whose brand markings did not match the supplier's records. On the desk stood a powered-on laptop with a card reader attached, holding an SD card from the packing-station camera that the site manager said had been formatted that morning. Both were seized; the case requires physical acquisition, imaging, preservation, carving where necessary and safekeeping.

Table 3.1: Known Facts, Hypothetical Scenario Data and Assumptions

Category	Content
Given by the brief	A laptop seized in a suspected data-theft-style investigation with an internal storage device and an attached USB device; collection must not modify the originals; the case requires physical acquisition, forensic imaging, preservation, data carving and safekeeping; IXImager and the ASB format are under consideration.

Category	Content
Hypothetical scenario data	Case id CT-2026-114, exhibits CT-E1 (laptop: Lenovo ThinkPad E15 Gen 3, 256 GB NVMe, NTFS, no volume encryption, Windows 11 Pro, powered on) and CT-E2 (64 GB SDXC card, exFAT, physical lock tab engaged, presented to the laptop through a generic USB mass-storage reader, which is the USB device of the brief); contents sought: 214 product photographs, Stock_Audit_Oct26.xlsx and three deleted shipment records; in-camera format on the morning of 6 Oct 2026.
Assumptions and exclusions	The seizure was lawful and recorded; the site lost network access when the laptop was placed in airplane mode; the camera reports no time zone; the laboratory has a hardware blocker with an SD adapter and a second reader. Supplier verification of the bearings, staff interviews and valuation of the goods are matters for the case officer, not the examiner.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 15–22)

2.1 Collecting and Transporting the Laptop and the Memory Device Before Imaging (Question 15)

Question 15: Prepare a step-by-step procedure for safely collecting and transporting the laptop and USB device before imaging.

Table 3.2: Step-by-Step Collection and Transport Procedure

Step	Action	Record produced
1. Secure and describe	Note the running state on arrival, photograph the desk including the reader and card in situ, and sketch cable runs before anything is touched.	Scene photographs with times
2. Preserve volatile state	Do not unplug the running laptop: shield the keyboard, connect to mains, capture memory and process data, then set airplane mode and radio-off. No shutdown before that capture is verified, since power-off may seal an encrypted volume.	Live-capture log, power-state note, RAM hash

Step	Action	Record produced
4. Remove and protect the card	Eject logically where the session permits, otherwise after power down; slide the physical write-protect tab to locked; handle by the edges, contacts up.	Card serial, lock-tab state
5. Bag separately	Laptop in an anti-static, shock-absorbing case; card in an antistatic sleeve, taped so it cannot slide out; each bag labelled at source with exhibit number, make, serial, seized-from, date and time, and sealed with a frangible evidence seal.	Seals CT-SEAL-4411/4412; custody form opened
7. Transport and intake	Travel direct in a hard case, out of a vehicle boot, away from magnets and heat; on arrival verify and photograph the seals, enter both items in the evidence register and store in a locked vault — imaging never happens at the scene.	Transport and intake entries (CUST-02)

Two of these steps are commonly done wrong: powering down a live laptop before volatile capture destroys evidence of open files and mounted volumes, and carrying a card loose in a bag risks physical damage or accidental writes from a host that auto-mounts it.

2.2 Purpose of Physical Acquisition and Why Analysis Uses an Image (Question 16)

> **Question 16: Explain the purpose of physical acquisition and why analysis should normally be performed on a forensic image rather than the original storage media.**

Physical acquisition copies the device sector by sector rather than file by file, capturing what a logical view cannot: deleted but unreallocated clusters, directory metadata, slack space, boot and reserve areas, and any partition the operating system no longer mounts. After an in-camera format that is the whole point — the shipment records exist only as clusters referenced by nothing.

- **Every mount writes.** Even an apparently read-only mount updates access times, a journal or a thumbnail cache, so the exhibit changes under examination.

- **A single copy has no recovery, and findings must be repeatable.** A faulty reader or a mistaken repair would end the case with nothing to re-examine, and a reviewing officer must be able to test the same question on the same bytes.
- **Custody stays continuous.** Each hand-off of a live exhibit invites an allegation of tampering; the original must survive in the condition in which it was seized.

2.3 Forensic Imaging Workflow Using IXImager (Question 17)

Question 17: Design a forensic imaging workflow using IXImager. Include source identification, write protection, imaging, verification, hashing, and documentation.

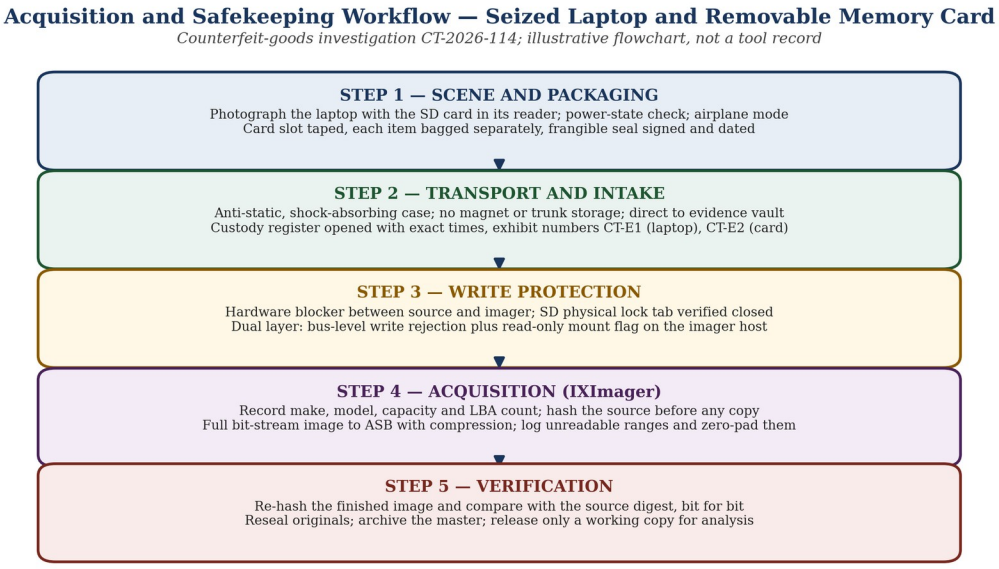


Figure 3.1: Acquisition and safekeeping workflow for the seized laptop and memory card (original illustrative flowchart, not a tool record)

Figure 3.1 sets out the five stages in the order they must happen; the table below expands the acquisition stage, which is the one the brief asks to be designed, and names the record each stage leaves behind.

Table 3.3: Imaging Procedure with the Record Each Stage Produces

Stage	What the examiner does	Recorded
Source identification	Read make, model, capacity, firmware and serial from the device itself, not the label; note LBA count and sector size so a truncated image is impossible to miss.	Device identity sheet

Stage	What the examiner does	Recorded
Write protection	Attach the hardware blocker between source and imager host; confirm the blocker reports write rejection; verify the card's lock tab is engaged and that the imager mounts read-only.	Blocker log, tab state
Acquisition	Acquire the full device to an ASB image, compression on, error ranges zero-filled and logged; image the laptop volume separately from the card.	Acquisition log with times
Verification	Re-read the finished image and compare its digest with the source digest computed before copying; on any mismatch, discard the image and repeat rather than repair it.	Verification entry, pass or fail
Hashing	Record SHA-256 for the source, the image and, later, the working copy, in the case register rather than only on screen.	Three-part hash record
Documentation	Record examiner, tool and version, profile used, clock offset and any anomaly; re-seal the originals and update the custody form.	Work notes, custody entry

2.4 Using Hash Values to Demonstrate Integrity (Question 18)

Question 18: Explain how hash values can be used to demonstrate integrity before and after acquisition.

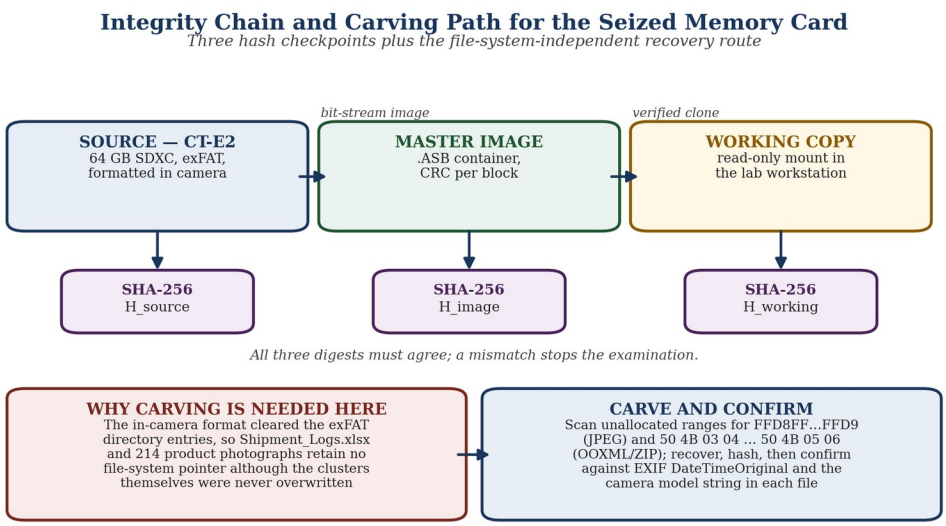


Figure 3.2: Three-checkpoint integrity chain and the carving path (illustrative diagram; digest values are held in the case register)

Figure 3.2 shows why three digests rather than one are recorded: the first fixes what was seized, the second proves the copy is faithful, and the third proves the analysis changed nothing.

Table 3.4: Hash Checkpoints in This Examination

Checkpoint	Computed on	What it proves	What it cannot prove
H1 before copying	The source device, read through the blocker	The exact bytes examined existed at seizure time	That those bytes were authored innocently
H2 after imaging	The ASB image re-read from disk	The image is a faithful copy of H1	Nothing about later handling of the original
H3 per working copy	The clone used by the analyst	Analysis did not alter the evidence set	Correctness of the analyst's interpretation

A hash proves only that content did not change between two measurements, yet that claim is load-bearing: without it, an assertion that a recovered invoice is the one the site deleted rests on the examiner's word alone. Values are held in the case register and abbreviated here as H1 to H3, since printing a digest without the register that produced it proves nothing.

2.5 Role of the ASB Container and Accompanying Information (Question 19)

Question 19: Describe the role of an ASB container in storing or managing forensic evidence and identify the information that should accompany the evidence.

The ASB format is a single-file container for a forensic image: it stores acquired sectors, permits compression for transport, records per-block integrity data so a damaged region can be located rather than merely noticed, and carries acquisition metadata beside the bytes. Its role is packaging and preservation at once — one object that can be hashed, moved, verified and mounted read-only, instead of a directory of loose segments whose completeness must be argued.

Table 3.5: Information That Must Travel With the Container

Field	Value in this case
Identity and acquisition	Case CT-2026-114; image of CT-E2 and a separate image of the CT-E1 volume; source capacity, file system, serial read from the device, lock-tab state, reader used; tool and version, examiner, start and end times, sector count, zero-filled error ranges
Integrity and timekeeping	Source and container digests with the algorithm named; per-block integrity reported as verified; imager clock source and its measured offset
Custody	CUST-01 and CUST-02 entries, seal numbers, vault location, and which working copies were derived, by whom and where held

2.6 Data Carving and When It Becomes Necessary (Question 20)

Question 20: Explain data carving and discuss when carving may be necessary if file-system information is unavailable or damaged.

Carving recovers files by recognising content instead of directory entries: a scanner walks raw ranges looking for a known header and, where the format supplies one, a matching footer or recoverable length. It is needed precisely when the file system can no longer point at data — after a format, after partition damage, on a file system the tools handle poorly, or where the index was overwritten but not the payload.

- **Photographs.** JPEG start FFD8FF and end FFD9, with the embedded thumbnail in the same file; complete images recover with EXIF intact, but carved file names are synthetic, so a mapping table is required.
- **Spreadsheets.** OOXML is a ZIP: local header 50 4B 03 04 confirmed by an end-of-central-directory record; a truncated archive loses its later sheets, which must be stated rather than quietly dropped.

- **Fragmented and unreadable ranges.** Scattered clusters may yield parts only, and where a read fails the ranges are recorded — no finding may rest on space that was never examined.

Two cautions govern how carving output is reported: a carved file's file-system times are gone, leaving only weaker content metadata for dating; and carving re-finds whatever the ranges hold, including files deleted for innocent reasons, so recovery is never itself evidence of concealment.

2.7 Date and Time Problems and Documenting Uncertainty (Question 21)

Question 21: Discuss date/time problems in digital evidence and explain how investigators should document time-zone, clock, and timestamp uncertainties.

Table 3.6: Time Problems Present in This Case and How Each Is Documented

Problem	How it arises here	Documentation rule
Zone not recorded	exFAT and EXIF DateTimeOriginal store local time with no zone	State the zone assumed and why (site at IST); never convert silently
Clock drift and skew	Laptop clock measured 2 min 40 s ahead of the case reference; camera unknown	Record each device's offset at acquisition and apply it before correlating
Set versus true time	A user may leave a wrong date; a photo can look two years early	Treat device time as asserted and corroborate against an external event
Precision mismatch	Camera to one second, NTFS to 100 ns, logs to one minute	Report the coarsest precision, and keep UTC canonical with IST shown beside it

The writing rule that follows is mechanical: every time in this report is presented as IST with the UTC value in the same cell, each source's offset is stated once in the method section, and any statement that depends on an ordering tighter than one minute is flagged as provisional rather than concluded.

2.8 Chain-of-Custody Record (Question 22)

Question 22: Prepare a chain-of-custody table for the seized laptop from collection through laboratory examination.

Table 3.7: Chain-of-Custody Form — Exhibits CT-E1 and CT-E2 (hypothetical completion)

Entry	Item	Released by	Received by	Date and time (IST)	Purpose and condition	Signatures
CUST-01	CT-E1 laptop; CT-E2 SD card in reader	Site manager (hypothetical)	SE-1 seizing officer	06 Oct 2026, 11:42	Seized at premises; laptop powered on, card lock tab engaged; sealed CT-SEAL-4411 / 4412	SE-1 / witness
CUST-02	CT-E1, CT-E2	SE-1	Evidence Custodian, CEL vault	06 Oct 2026, 13:05	Transport direct in hard case; seals photographed on arrival and intact	SE-1 / Custodian
CUST-03	CT-E2	Custodian	EX-2 examiner	07 Oct 2026, 09:20	Issued for write-blocked acquisition; resealed on return 16:40 the same day	Custodian / EX-2
CUST-04	Image ASB-CT-E2-01; CT-E1	EX-2 / Custodian	Exhibit register; EX-2 examiner	07 Oct 11:55; 08 Oct 10:10	Image created, hashed, verified and held read-only with its working copy; laptop issued for volume image with seal intact	EX-2 / Recorder / Custodian
CUST-06	CT-E1, CT-E2	EX-2	Custodian	10 Oct 2026, 17:30	Returned after examination; seals CT-SEAL-4413 applied; report pending	EX-2 / Custodian

A custody record is worth anything only if it is continuous: every minute an exhibit exists must be owned by a named holder, and a gap is a question the examiner cannot answer later. The form therefore records condition and seal state at each movement, and enters the image as an exhibit of its own so its derivative status is explicit.

3. CONSOLIDATED INTEGRITY STATEMENT AND CONCLUSION

For a matter turning on photographs and shipment records, the examination's value rests on process rather than cleverness. The card was seized locked, acquired through a hardware blocker, hashed before and after, wrapped in an ASB container carrying its own integrity data, and never analysed as an original. After the in-camera format the records were recovered by content rather than by name, each with its provenance and the acknowledged loss of file-system times. Times are stated in IST against a named UTC reference with the laptop's measured 2 min 40 s offset applied. Within those bounds the examination supports three statements: the card held images consistent with goods bearing the supplier's marks; those images and the recovered spreadsheets survived the morning format; and the file system no longer referenced them, which is consistent with either a camera format or a manual deletion. Telling those apart is beyond the media alone and is reported as an open question.

4. REFERENCES

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ABSTRACT

This report answers Questions 23 to 30 for a hypothetical examination of DGN-FS02, a design-department file workstation whose Skyline_Residences review package was found missing two days before an internal compliance review. A forensic image of the volume is available and Volume Shadow Copies exist, so earlier versions can be recovered and compared. The four required deliverables are produced: an evidence classification table covering user-created, system and application, suspicious, deleted or recovered and contextual material; an event timeline for the document lifecycle; a Volume Shadow Copy recovery and comparison workflow; and a validation checklist that a recovered artefact must pass before it becomes a finding. The scenario is deliberately built around a benign mechanism — a document-management client that relocates working files — to show why recovery proves state at a point in time and never proves who deleted anything. No tool was executed for this coursework; all values are scenario data.

CONTENTS

1.	Case background, facts and assumptions
2.	Detailed answers to Questions 23 to 30
3.	Conclusion and stated limits
4.	References

1. CASE BACKGROUND, KNOWN FACTS AND ASSUMPTIONS

The college's design studio runs an annual internal compliance review of sponsored student projects. For the Skyline Residences package, the reviewer opened the project folder on 3 October 2026 and found the calculation note, the approval tracker and the staircase detail drawing absent, with no entry in the tracker for the review cycle. The department's IT support had configured a nightly backup agent on DGN-FS02, which creates Volume Shadow Copies, and a ProjectVault document-management client on the drafters' machines. A forensic image of the volume was taken before anyone attempted a restore.

Table 4.1: Known Facts, Hypothetical Scenario Data and Assumptions

Category	Content
Given by the brief	A person suspected of deleting documents and browser-related artefacts after unauthorised activity; a forensic image of a Windows system is available; Volume Shadow Copies are present; deleted or changed artefacts may be recoverable; the team must select relevant evidence, remove irrelevant data, analyse artefacts, validate findings and reconstruct the sequence.
Hypothetical scenario data	Case id DGN-2026-07; DGN-FS02 (Windows 11 Pro, NTFS, 1 TB, shadow storage on the same volume); ShadowCopy1 captured 1 Oct 18:00 and ShadowCopy2 captured 2 Oct 18:00 IST; project folder D:\Design\Skyline; files Skyline_Calc_Note.docx, Approval_Tracker.xlsx, Skyline_Staircase_Detail.dwg and the Revit model; archive path \\DGN-FS01\Archive\Skyline; compliance review scheduled 5 Oct 2026.
Assumptions	The image is verified against its acquisition hash; the backup agent's schedule is documented and was not altered; the ProjectVault client is a college-licensed tool with server-side sync records that the review cannot compel without the vendor's co-operation.
Interpretive stance taken	The absence of a file from the live folder is treated as an observation to be explained, not as misconduct. Alternative mechanisms are tested before any deletion is asserted.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 23–30)

2.1 Selecting and Categorising Evidence Before Detailed Analysis (Question 23)

Question 23: Explain how investigators should select and categorize digital evidence before detailed analysis.

The image of a file server holds roughly 260,000 files, of which perhaps 900 bear on four days of one project. Selection is therefore the first analytical act, and it is done by bounding rather than by browsing, so that the reduction is itself reviewable.

- **Bound by time.** Keep only artefacts whose recorded times fall inside 30 Sep to 3 Oct 2026, widened by the measured clock uncertainty of the volume, plus anything the snapshot boundaries freeze.
- **Bound by place.** Retain the project folder and its archive twin, the volume's metadata files, the shadow storage, the user hives of the three drafters, the backup agent's logs and the DMS client cache.
- **Bound by type.** Documents, drawings and models first; executables and libraries only where they are the producer of an artefact, such as the Office binaries that write the metadata being examined.
- **Reduce with known-file lists.** An NSRL comparison removes operating-system and application binaries; each removal is recorded by rule, not by silent exclusion, so the reviewer can re-include.
- **Categorise as you go.** Every retained item receives a class from the scheme in Question 24 at the moment it is retained, together with the exhibit and the snapshot or live path it came from.
- **Separate the irrelevant visibly.** Material judged irrelevant is moved to a quarantine list with the reason, which is what lets a third party check that nothing important was dropped.

2.2 A Classification Scheme for the Retained Evidence (Question 24)

Question 24: Prepare a classification scheme for user-created files, system/application files, potentially suspicious files, deleted/recovered artifacts, and supporting/contextual evidence.

Table 4.2: Five-Class Scheme Applied to the Skyline Package

Class	Definition applied	Examples in this case	Handling rule
C1 User-created	Content authored or edited by a person through an application	Skyline_Calc_Note.docx, Approval_Tracker.xlsx, Skyline_Staircase_Detail .dwg	Preserve original order of content; hash each version separately; never edit for legibility
C2 System and application	Produced by the operating system or an application as a by-product	$MFT, $UsnJrnl, Event Logs, Office jump lists, backup-agent schedule log	Read as machine record; check service and task context before interpreting a time

Class	Definition applied	Examples in this case	Handling rule
C3 Potentially suspicious	Behaviour that is unusual in this environment and needs explanation	Bulk rename of the working folder at 2 Oct 11:04; archive copy with created later than modified	State the anomaly, then test the ordinary explanation before any adverse inference
C4 Deleted or recovered	Absent from the live tree, present in a snapshot or in unallocated space	v1 of the calculation note in ShadowCopy1; pre-removal tracker rows in ShadowCopy2	Label with the exact snapshot; do not present a recovered file as a live one
C5 Contextual and supporting	Explains the environment rather than the act	Review calendar entry, DMS sync configuration, roster of project members, folder ACLs	Use to bound alternatives; keep out of findings as stand-alone proof

The scheme is deliberately orthogonal: one artefact may hold two tags, for example a recovered file (C4) that is also user-created (C1). Tags answer how an item is handled; they do not by themselves answer what it means.

2.3 How Event Analysis Reconstructs a Sequence (Question 25)

Question 25: Explain how Event Analysis can assist in reconstructing a sequence of activities.

Event analysis means taking the timestamped records the system produces on its own and ordering them so that the gaps become as informative as the entries. Four sources matter on this volume, and each answers a different question about the same four days.

- **NTFS $MFT.** Per-file created, modified, accessed and entry-changed times, plus the record sequence number — the source that shows whether a file's directory entry itself was rewritten.
- **$UsnJrnl.** A change journal of create, data extend, rename and close events; it preserves the order of operations on a file even after the file is gone from the live tree.
- **Security and application event logs.** Who opened a file, from what session, and when a task or service ran — the register that separates a person's logon from an agent's schedule.
- **Application and agent logs.** The backup agent's snapshot times and the DMS client's sync notices, which is where a relocation reveals itself as a routine event rather than a deletion.

Ordering these against one another produced the pivotal observation here: at 2 October 11:04 the journal records a rename of the folder and a create on the archive path within two seconds of each other, and no delete record exists for the individual files. That pattern is what a client-initiated relocation looks like; a deletion by a person in Explorer does not write the archive create.

2.4 Purposes of Cloud Analysis and Lead Analysis (Question 26)

Question 26: Explain the purpose of Cloud Analysis and Lead Analysis in an investigation and identify the kinds of questions each may help answer.

Table 4.3: Two Different Functions, Both Needed Before Conclusions

	Cloud analysis	Lead analysis
Purpose in this case	Obtain the off-host record of the same events: the DMS service's sync log, the share's own access log and the review calendar held on the college tenant.	Keep an organised list of unresolved questions so that each thread is either followed to an artefact or closed with a reason.
Questions it answers	Did the archive copy arrive from the client or from a server-side policy? What did the tracker look like at 11:04, from a source the workstation cannot rewrite?	Which file did the tracker say was missing? Whose session touched the folder? Why does no delete record exist? What would falsify the relocation explanation?
Evidence types it yields	Provider-side audit records, object identifiers, retention limits and timestamps from a clock the suspect volume cannot influence.	Hypotheses to test, artefacts still unexamined, sources to request, and the list of what cannot be obtained at all.
Principal limit	Availability depends on provider retention and on authorisation the department does not control.	Leads are not findings; the risk is that a promising thread hardens into an assertion.

2.5 Volume Shadow Copy Analysis for Recovery and Comparison (Question 27)

Question 27: Describe how Volume Shadow Copy analysis can help recover or compare earlier versions of evidence.

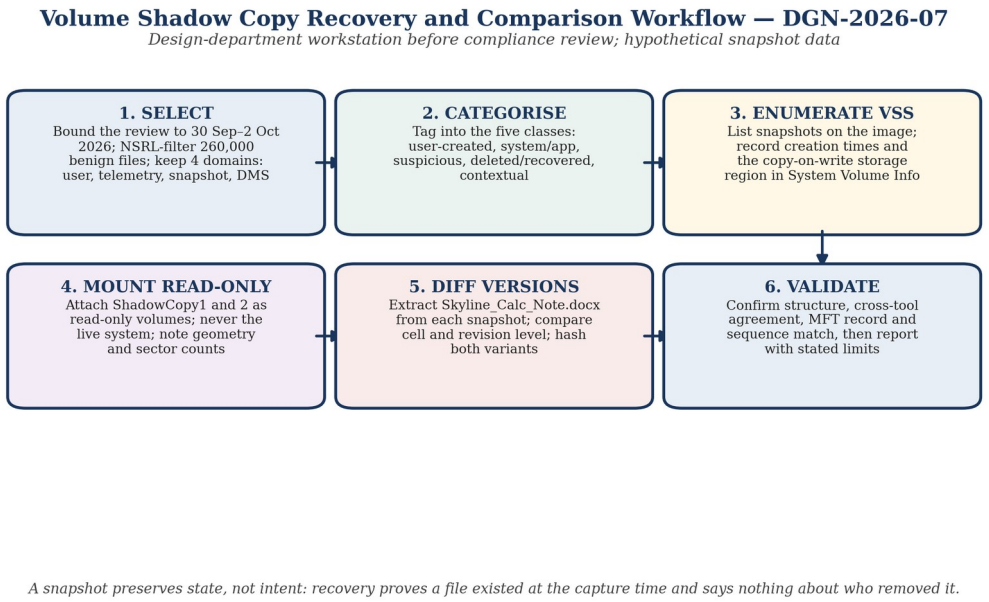


Figure 4.1: Shadow copy selection, mounting, comparison and validation workflow (illustrative diagram, not a tool screenshot)

Figure 4.1 shows the six stages, the last two of which are where a recovery is usually over-read: a recovered file is not a live one. A shadow copy is a point-in-time view of the volume, so it can answer a question the live file system cannot: what did this path hold at 1 October 18:00? Comparison is the more valuable use, because the difference between two snapshot views is an observation about change rather than about loss. In this case the calculation note recovered from ShadowCopy1 differs from the version in ShadowCopy2 by three revision blocks, which is consistent with the review-cycle edits described by the drafters, and the tracker rows that the reviewer could not find are present in the second snapshot with their original times.

Two mechanical points decide whether this is usable. Shadow storage must be read from the image, never from the live system, so that the analyst's own activity cannot be confused with case activity; and each recovered file must be labelled with the snapshot it came from, because a recovered v1 and a live v2 are different exhibits even when the file names match.

2.6 A Hypothetical Lifecycle: Create, Modify, Copy, Delete, Recover (Question 28)

Question 28: Construct a hypothetical timeline showing how a file could be created, modified, copied, deleted, and later recovered from a Volume Shadow Copy.

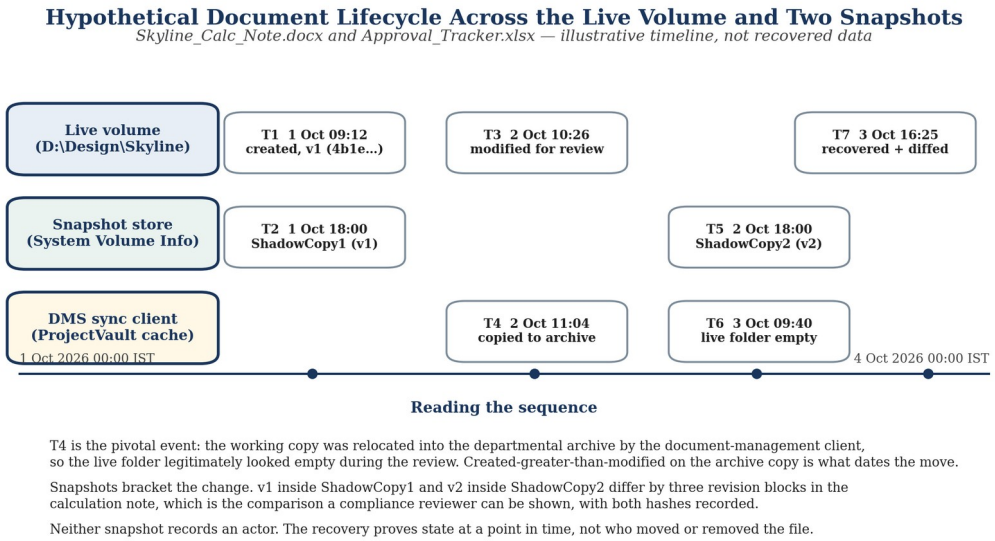


Figure 4.2: Document lifecycle across the live volume and two snapshots (hypothetical scenario data)

Figure 4.2 places those moments on three rows - the live volume, the snapshot store and the document-management client - so that the reader can see which record source answers which claim.

Table 4.4: Hypothetical Lifecycle of Skyline_Calc_Note.docx

Ref	Time (IST)	Event	What the record shows
T1	1 Oct 09:12	Created by a drafter in the project folder	MFT created equals modified; $UsnJrnl CREATE
T2	1 Oct 18:00	Backup agent writes ShadowCopy1	Snapshot freezes v1 with its original times
T3	2 Oct 10:26	Modified during the review cycle	DATA_EXTEND plus an Office jump-list entry under that profile
T4	2 Oct 11:04	Copied to the archive path, then removed from the live folder	Created later than modified on the archive copy; rename of the parent recorded, no per-file delete
T5	2 Oct 18:00	ShadowCopy2 written	Holds v2, the archive copy and the tracker rows as they stood
T6	3 Oct 09:40	Image acquired for the review	Live folder empty; reviewer sees nothing

Ref	Time (IST)	Event	What the record shows
T7	3 Oct 16:25	Examiner mounts both snapshots, recovers and diffs	Two labelled versions, each hashed; relocation confirmed by the archive pair

2.7 Validating a Recovered Artefact Before It Becomes a Finding (Question 29)

Question 29: Explain how the recovered artifact should be validated before it is included as a finding in the final report.

Table 4.5: Validation Checklist for Recovered Artefacts

Check	What must be true	Status here
VAL-01 Provenance	The snapshot and path of origin are recorded and unambiguous	Pass — ShadowCopy1, D:\Design\Skyline
VAL-02 Structure	The file opens as its extension claims, with an intact internal structure	Pass — valid OOXML package
VAL-03 Integrity	A digest is recorded for the recovered object at recovery time	Pass — logged in the work notes
VAL-04 Independence	A second reader or tool reaches the same result	Pass — two parsers agree on revision count
VAL-05 Cross-source match	Metadata agrees with an independent record, not only with itself	Pass — archive copy present at T4 in the share log
VAL-06 Time sanity	Timestamps are consistent with the measured clock offsets and the snapshot boundary	Pass within the 2 Oct 11:04 window
VAL-07 Alternative test	A benign mechanism has been considered and either excluded or retained	Retained — relocation explains the pattern
VAL-08 Expression	The finding states what the artefact shows and stops there	Pass — no attribution to a person is claimed

The order is not cosmetic. Structure and integrity are checked before meaning is argued, and an artefact that fails VAL-05 or VAL-07 is reported as an observation rather than a finding, however tempting its content looks.

2.8 Sources of Error and Alternative Explanations (Question 30)

Question 30: Identify possible sources of error or alternative explanations that could affect the interpretation of timestamps and deleted artifacts.

- **Timestamp copying and inheritance.** A relocated file can carry an older modified time with a new created time; reading that pair as a deletion window is a familiar and serious error.
- **Journal and snapshot lag.** $UsnJrnl entries are written by the volume manager, and a snapshot taken minutes after an event may not contain what the reader expects.
- **Restore-point noise.** Installers, antivirus updates and system protection can create snapshots that were never intended as evidence points, so their trigger must be identified.
- **Shadow storage truncation.** When the allocated area is exceeded, the oldest copy is deleted; an absent version may mean 'not retained' rather than 'never existed'.
- **Tool artefacts.** Reparse and extraction steps can rewrite access times, and a mount that was not read-only leaves the analyst's own activity in the record.
- **Benign alternatives.** Routine archiving, a scheduled cleanup policy, a sync conflict, a user moving files to a new project folder, or a file that was never in this path at all.

3. CONCLUSION AND STATED LIMITS

The package was not destroyed. Two shadow copies and one archive path together show that the calculation note, the tracker and the drawing were relocated into \\DGN-FS01\Archive\Skyline at 11:04 on 2 October, that the review-cycle modifications are present between ShadowCopy1 and ShadowCopy2, and that the tracker rows the reviewer could not find exist in the second snapshot with their original times. What the artefacts cannot do is name a person or state an intention: the shadow copies record the volume's state at a capture time, the sync client's activity is indistinguishable from a user's within the journal alone, and the DMS service log that would settle which one moved the files sits with the vendor under a retention period the department has not checked. The report therefore presents a recovery finding, a relocation finding and a recommendation — that the compliance review be reopened on the archive copies, and that snapshot triggers and retention be documented by policy so that a future reviewer is not obliged to reason from a rename record to a conclusion about a person.

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ABSTRACT

This report answers Questions 31 to 38 for a hypothetical incident in which the project workspace of a student team on a fictional collaboration service, TeamForge, was renamed, stripped of two pages, had a design document exported externally and received an abusive guest post, all within thirty minutes under one account's credentials. Six authorised evidence domains are available: a Windows laptop, its browser data, mail records, a messaging export, an intern's Android handset and the provider's audit log. The four required deliverables are produced — a multi-source evidence map, a cross-source correlation matrix, an analysis of the challenges that distributed evidence creates, and a final investigative summary that separates direct from corroborating from unresolved matters. The thread that binds the sources is an OAuth refresh token and a session identifier, not a username, and that is what lets the examination clear the account holder while leaving the person who held the tablet unidentified. No tool was executed for this coursework; all values are scenario data.

CONTENTS

1.	Case background, facts and assumptions
2.	Detailed answers to Questions 31 to 38
3.	Final investigative summary
4.	References

1. CASE BACKGROUND, KNOWN FACTS AND ASSUMPTIONS

On 29 September 2026 the team behind a fourth-year project noticed that its workspace pages had been retitled, a shared design document had been exported as a PDF by an unfamiliar client, and a message had been posted in their project channel under the name of the team lead, p_raman. The account's own owner says he was in a hostel room that evening with his laptop closed. The college's IT cell obtained the account owner's written consent for the laptop and mail, the intern consented for her phone, the team consented for the messaging export, and the department supplied the tenant gateway logs. Nothing was examined without an authorisation on file.

Table 5.1: Known Facts, Hypothetical Scenario Data and Assumptions

Category	Content
Given by the brief	A student account suspected of being used in a cyber incident involving unauthorised access to an online service; authorised access to a Windows computer, browser artefacts, e-mail messages, messaging records, a mobile phone and relevant internet or cloud records; evidence distributed across sources, to be correlated while weighing the challenges of modern digital forensics.
Hypothetical scenario data	Case id IDO-2026-19; incident window 28 Sep 2026 18:02–18:31 IST; account p_raman in workspace SKY-STUDIO; exhibits IDO-E1 laptop (Windows 11, NTFS), IDO-E2 its browser profile, IDO-E3 mail records, IDO-E4 messaging export, IDO-E5 intern's Android handset (Moto G32, FBE enabled), IDO-E6 provider audit log and tenant gateway records; token id TK-77c1; session id SS-4e19; source address 49.40.x.x attributed to a hotel gateway behind a VPN exit; client string curl/8.5.0 on Linux for the export.
Assumptions	Consents are documented for every device; the provider retains audit records for 30 days and object content for 7; workstation, handset and provider clocks differ by up to 90 seconds; the shared studio tablet has no badge or camera coverage of the desk it sits on.
Out of scope	Any finding about the intern's conduct, the content of the exported document, and disciplinary or legal characterisation of the impersonation.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 31–38)

2.1 Potential Evidence from Windows and Other Operating Systems (Question 31)

Question 31: Identify the potential evidence available from Windows and other operating systems in this case.

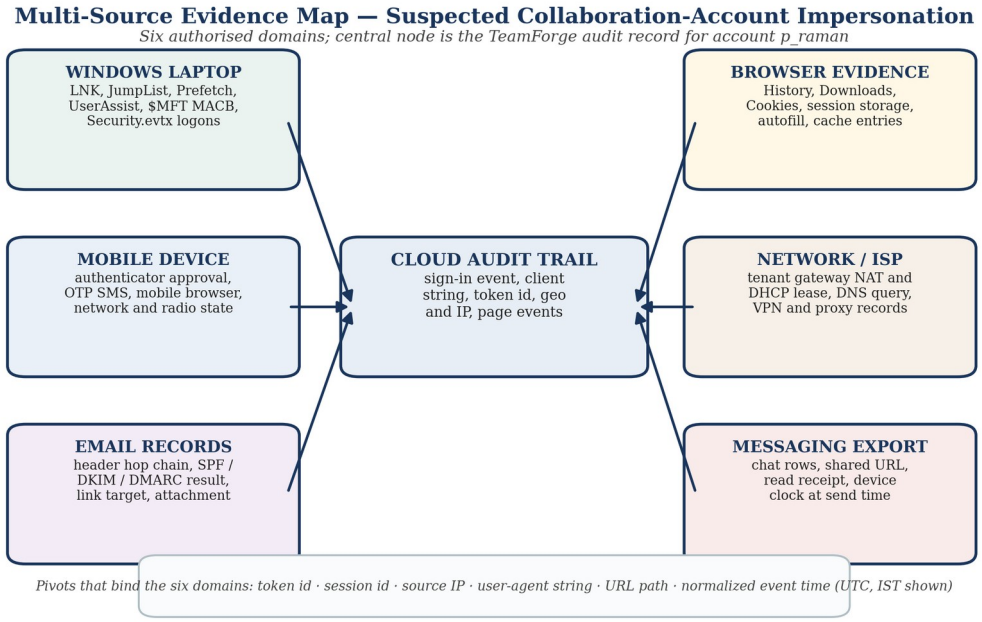


Figure 5.1: Multi-source evidence map centred on the provider audit trail (illustrative diagram)

Figure 5.1 maps the six authorised domains against the provider record at the centre; the arrows are the pivots actually available in this case, and the strip beneath names each one.

Table 5.2: Evidence by Operating Environment

Environment	Artefacts of use here	Question it answers
Windows 11 laptop (IDO-E1)	LNK targets, JumpLists, UserAssist, Prefetch, $MFT MACB times, Security.evtx logons, browser databases, Wi-Fi profiles	Whether the account holder's machine held a session in the window at all
Android handset (IDO-E5)	Authenticator approvals, SMS OTP, browser history, notification log, network and radio state, app usage stats	Which device actually received and answered the multi-factor prompt
Browser profile on each host	History, downloads, cookies, session storage, autofill, cache entries	Whether the phishing link was opened, and from what profile
Provider and gateway logs (IDO-E6)	Sign-in events with client string and geo, token issuance, object events, NAT and DHCP leases, DNS queries	Where the session came from and what the token did once issued
Messaging and mail stores (IDO-E2 to E4)	Message headers with hop chain, SPF and DKIM results, chat rows carrying the shared link, read receipts	How the credential was approached, and who was addressed

The Linux side of the incident — the export performed with a command-line client — has no host inside the college to examine, which is itself the finding: two of the six domains describe an actor the examiner can never reach, and the report must reason about the token rather than about a machine.

2.2 What Browser Evidence Reveals and Why It Needs Care (Question 32)

> **Question 32: Explain what browser artifacts may reveal about user activity and why browser evidence must be interpreted carefully.**

Browser data is the closest thing to a recording of attention: a history entry says a page was visited, a download row says bytes were saved to a path, and session storage can carry the address of the page that redirected the user. It is also the most fragile evidence in the case.

- **What it can show.** Visits with second-level timestamps, referrer chains, the destination of a typed or clicked link, files saved and the directory they were saved into, and sign-in failures.
- **Why it is fragile.** One cleared-browsing-data action, one private window, or one profile synced from another device can remove or import whole intervals without leaving an obvious trace in the history database itself.
- **Timestamp semantics.** Chromium-era stores count microseconds since 1601 UTC; a wrong conversion is a four-hundred-year error, and the local zone is not stored with the value.
- **Inference over presence.** A cached asset proves the browser fetched it, not that a person read a page; an autofill entry proves a field was filled once, not by whom.
- **Cross-checks that survive.** Download rows usually agree with an NTFS creation record, and a synced profile agrees or disagrees with the server-side session log — both comparisons are reported rather than assumed.

2.3 Evidence from Email and Messaging Systems (Question 33)

> **Question 33: List the types of evidence that may be available from email and messaging systems and explain how they can be correlated with other artifacts.**

Mail and messaging records supply three things no other source in this case contains: the approach to the credential, the exact link that was served, and a chain of server receipts that a workstation cannot alter.

- **Message headers and hop times.** Received lines give each relay a timestamp; the first hop places the sending host, and the gap between hops is a check on forgery.
- **Authentication results.** SPF, DKIM and DMARC verdicts say whether the sending domain vouched for the message; here the lure passed SPF and carried no DKIM signature, which is the classic shape of a look-alike sender.
- **Link target and attachment.** The displayed text and the underlying address differ; the extract of the target is what is cited, not the visible label.
- **Messaging rows and delivery state.** A chat entry carrying the same short link, with its send time and read receipt, is what ties a message to a device session.
- **Correlation rule applied.** Each of these is matched to the browser history entry and to the provider sign-in event only when all three fall inside the measured clock uncertainty; otherwise the relationship is reported as temporal proximity, not as a single chain.

2.4 Forensic Considerations and Challenges for the Mobile Phone (Question 34)

Question 34: Explain the forensic considerations when examining a mobile phone and identify challenges associated with mobile evidence.

- **Consent and separation of data.** The handset is the intern's personal property; extraction is limited by the consent text to the incident window and to the two applications named in it.
- **Encryption and lock state.** File-based encryption plus a screen lock means an unlocked acquisition at the scene is worth more than a locked one; if locked, the passcode route is documented rather than circumvented.
- **Logical versus filesystem scope.** A logical extract returns live application data; deleted chat rows and database write-ahead logs need a filesystem-level image, which this consent did not extend to.
- **Application containers and retention.** Authenticator approvals are held by the authentication app, whose own log retention may be days; a notification history can be shorter still.
- **Clock and network context.** Handset time is set by the network, so it can lead a laptop by tens of seconds; radio and Wi-Fi state is what shows the device was on cellular, away from the studio.

2.5 Internet and Cloud-Based Evidence Sources (Question 35)

Question 35: Discuss Internet and cloud-based evidence sources that may be relevant to the investigation.

Table 5.3: Off-Host Sources Considered

Source	What it could add	Availability in this case
Provider audit trail	Sign-in events, token issuance, client strings, object renames, exports	Available for 30 days, event metadata only
Provider object content	The content of the exported design document	Retention of 7 days had lapsed before the report — unavailable
Tenant gateway and DHCP	Which college host held the NAT address at 18:11	Available for 90 days; rules out an on-campus origin
DNS resolvers	Which names the laptop resolved that evening	Available; no resolution of the lure host from the account holder's machine
VPN or proxy exit	Who used the exit address	Requires provider co-operation and is outside the department's authority
Third-party authenticator	Approval timestamps and the device that answered	Available only through the account owner's own app data

Cloud sources differ from the others in kind, not merely in location: their retention, format and availability are set by a contract the college did not negotiate, and the examiner's window to ask is measured in days.

2.6 Cross-Source Correlation Matrix (Question 36)

Question 36: Prepare a cross-source correlation table linking at least five hypothetical artifacts from the computer, browser, email/messaging, mobile, and cloud sources.

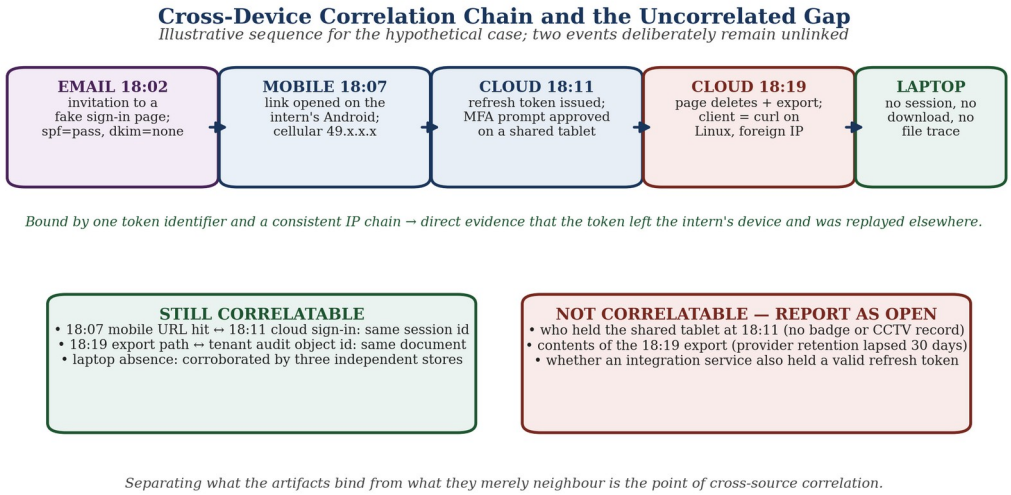


Figure 5.2: Correlation chain across devices and the deliberately unresolved gap (illustrative)

Table 5.4: Correlation of Six Artefacts Across Five Sources

Artefact (source)	Time (IST)	Correlating value	What the link establishes	Confidence
Lure message (IDO-E3 mail)	28 Sep 18:02	Target URL; sender domain	The link the browser later opened existed in the mailbox	High
Mobile browser visit (IDO-E5)	18:07:41	Same URL, same device session	The link was opened on the intern's handset, not the laptop	High
Authenticator approval (IDO-E5)	18:10:55	Session id SS-4e19	An approval was answered on a device other than the owner's	High
Provider sign-in (IDO-E6)	18:11:02	SS-4e19; TK-77c1; 49.40.x.x	The token issued to that approval was replayed from a foreign address	High
Messaging row (IDO-E4)	18:06:58	Short link hash	The same lure circulated in the team channel within a minute of the mail	Medium
Laptop absence (IDO-E1/E2)	18:00–18:35	No logon, no history row, no download	The account holder's machine did not host the session	Medium-high, absence-based

Figure 5.2 draws the same chain as a sequence, with the unresolved gap left visible. The matrix is read down the correlation-value column rather than across the rows: two identifiers, SS-4e19 and TK-77c1, travel from the handset to the provider and bind four sources without

any reference to the account name. Absence evidence in the last row is graded lower because silence in a store proves only that the store holds nothing.

2.7 Challenges When Evidence Is Distributed (Question 37)

Question 37: Explain at least six major challenges in digital forensics when evidence is distributed across multiple devices and cloud services.

Table 5.5: Six Challenges and the Response Adopted Here

Challenge	How it shows in this case	Response
Clock skew and time zones	Handset, laptop and provider differ by up to 90 seconds; stores hold different precisions	Normalise to UTC, record each offset, report a tolerance with every link
Identifier drift	The account name is on everything and means nothing; the session id is on four items	Pivot on tokens and sessions, never on usernames
Retention asymmetry	The laptop keeps months, the messaging export weeks, the provider 30 days and object content 7	Ask the shortest-retention source first and record each expiry as it passes
Jurisdiction and authority	The VPN exit and the foreign cloud sit outside the department's reach	Document what was requested, from whom and under what authority; report the refusal as a limit
Volume and format heterogeneity	Six stores, five formats, three of them proprietary	Reduce with known-file filtering, then convert into one timeline with source, value and confidence fields
Plausible innocence at every hop	An intern clicking a link, a shared tablet approving a prompt, a team export routine	Test each benign explanation explicitly before grading any artefact as suspicious
Attribution to a person	Devices are identified; the hands holding them are not	State conclusions about sessions and tokens, and leave person-level claims to corroboration outside the data

2.8 Final Investigation Conclusion (Question 38)

Question 38: Prepare a final investigation conclusion distinguishing direct evidence, corroborating evidence, and unresolved questions.

Table 5.6: Three Grades of Conclusion

Grade	Matters placed in it
Direct evidence	Session SS-4e19 was opened on the intern's handset at 18:07 and approved at 18:10:55; token TK-77c1 was issued to that session and used from 49.40.x.x with a curl client to rename two pages, delete one and export the design document.
Corroborating evidence	The lure arrived by mail at 18:02 and in the channel at 18:06; the account holder's laptop shows no session, no history row and no download in the window; the tenant gateway confirms no campus host held that address.
Unresolved questions	Who physically answered the prompt on the shared studio tablet; what the exported document contained, since provider retention lapsed; whether an integration service also held a valid refresh token; whether more than one person used the intern's handset between 18:02 and 18:11.

3. FINAL INVESTIGATIVE SUMMARY

The distributed evidence supports one coherent account and no more. A look-alike message reached two team members within four minutes; one of them opened it on a personal handset, approved a sign-in prompt that the service had routed to a shared studio tablet, and a token minted from that approval was then used from outside the college network to rename pages, delete one and export a design document under the team lead's account. The account owner's machine is excluded from the session by three independent stores. The chain is bound by a session identifier and a token identifier rather than by any username, and it stops short of a person at both ends: the intern's device, the tablet's approver and the external client belong to people this evidence cannot name. Two practical consequences follow for the department. Multi-factor prompts should never be routed to a shared device, because the approval is then unattributable; and because the provider's object content expired in seven days, a standing request template and a named contact should exist before the next incident, since in a distributed-evidence case the loss of a source is generally a loss of time rather than a loss of skill.

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